

# Big Basket-E-commerce-Database-Management-System

## Chapter 2

### 2.1 Introduction:

The entities required for the big Basket E-Commerce Database are identified based on grocery shopping requirements. The attributes of each entity along with their Primary Keys (PK) and Foreign Keys (FK) are defined to maintain proper data organization. The relationships between entities are analysed using cardinality and participation constraints. This design helps in creating the ER diagram, relational schema, and database implementation while ensuring data consistency and reducing redundancy.

### 2.2 Entity and their attributes and keys:

#### i. Entity: Customer(Strong Entity)

| Attribute     | Key Type    |
|---------------|-------------|
| Customer_ID   | Primary Key |
| Customer Name | Not Null    |
| Email         | Unique Key  |
| Phone No      | Unique Key  |
| Date of Birth | -           |
| Address       | -           |

#### ii. Entity: Category(Strong Entity)

| Attribute     | Key Type    |
|---------------|-------------|
| Category_ID   | Primary Key |
| Category Name | Unique Key  |
| Description   | -           |

#### iii. Entity: Seller(Strong Entity)

| Attribute   | Key Type    |
|-------------|-------------|
| Seller ID   | Primary Key |
| Seller_Name | Not Null    |
| Contact_No  | -           |
| Address     | -           |
| Email       | -           |

**iv. Entity: Product(Strong Entity)**

| Attribute    | Key Type    |
|--------------|-------------|
| Product_ID   | Primary Key |
| Product Name | -           |
| Category_ID  | Foreign Key |
| Seller_ID    | Foreign Key |
| Brand        | -           |
| Price        | Check       |
| Stock        | Check       |
| Expiry_date  | -           |

**v. Entity: Cart(Strong Entity)**

| Attribute    | Key Type    |
|--------------|-------------|
| Cart_ID      | Primary Key |
| Customer_ID  | Foreign Key |
| Total_items  | -           |
| Total_amount | -           |

**vi. Entity: Cart\_Item(Weak Entity)**

| Attribute    | Key Type    |
|--------------|-------------|
| Cart_Item_ID | Primary Key |

|            |             |
|------------|-------------|
| Cart_ID    | Foreign Key |
| Product_ID | Foreign Key |
| Quantity   | Check       |

**vii. Entity: Order(Strong Entity)**

| Attribute    | Key Type    |
|--------------|-------------|
| Order_ID     | Primary Key |
| Customer_ID  | Foreign Key |
| Order_date   | -           |
| Order_status | Check       |
| Total_amount | -           |

**viii. Entity: Order\_Item(Weak Entity)**

| Attribute     | Key Type    |
|---------------|-------------|
| Order_item_ID | Primary Key |
| Order_ID      | Foreign Key |
| Product_ID    | Foreign Key |
| Quantity      | -           |
| Price         | -           |

**ix. Entity: Payment(Strong Entity)**

| Attribute      | Key Type    |
|----------------|-------------|
| Payment_ID     | Primary Key |
| Order_ID       | Foreign Key |
| Payment_method | -           |
| Payment_status | Check       |
| Payment_date   | -           |
| Amount         | -           |

**x. Entity: Review(Strong Entity)**

| Attribute   | Key Type    |
|-------------|-------------|
| Review_ID   | Primary Key |
| Customer_ID | Foreign Key |
| Product_ID  | Foreign Key |
| Rating      |             |
| Review_text | -           |
| Review_date | -           |

**xi. Entity: Delivery(Strong Entity)**

| Attribute        | Key Type    |
|------------------|-------------|
| Delivery_ID      | Primary Key |
| Order_ID         | Foreign Key |
| Delivery_address | -           |
| Delivery_date    | -           |
| Delivery_status  | Check       |

**xii. Entity: Inventory(Strong Entity)**

| Attribute      | Key Type    |
|----------------|-------------|
| Inventory_ID   | Primary Key |
| Seller_ID      | Not Null    |
| Product_ID     | Not Null    |
| Stock_Quantity | Default     |
| Stock_Status   | Default     |
| Last_Updated   | Not Null    |

## 2.3 Entity Relationships

| Entity 1 | Relation    | Entity 2   | Cardinality  | Participation   |
|----------|-------------|------------|--------------|-----------------|
| Customer | Owns        | Cart       | One-to-One   | Partial/Total   |
| Customer | Place       | Order      | One-to- Many | Partial/Total   |
| Customer | Makes       | Payment    | One-to-Many  | Partial/Total   |
| Customer | Writes      | Review     | One-to-Many  | Partial/Total   |
| Category | Categorizes | Product    | One-to-Many  | Total/Total     |
| Seller   | Supplies    | Product    | One-to-Many  | Partial/Total   |
| Product  | Include-in  | Order_item | One-to-Many  | Partial/Total   |
| Order    | Contains    | Order_item | One-to-Many  | Total / Total   |
| Order    | Generates   | Payment    | One-to-One   | Total / Total   |
| Order    | Generates   | Delivery   | One-to-One   | Total / Total   |
| Product  | Receives    | Review     | One-to-Many  | Partial /Total  |
| Cart     | Contains    | Cart_Item  | One-to-Many  | Total / Total   |
| Product  | Included-in | Cart_Item  | One-to-Many  | Partial / Total |
| Seller   | Manages     | Inventory  | One-to-Many  | Partial / Total |
| Product  | Has         | Inventory  | One-to-Many  | Partial / Total |